

Radio Astronomy - Molecular Transitions

- Molecular transitions
 - Rotational Transitions
 - Diatomic Molecule
 - Energy Levels and Selection Rules
 - Radiative Transfer and Line Intensities
 - Partition Function and Column Density
 - Symmetric Top Molecule
 - Asymmetric Top Molecule

Radio Astronomy - Molecular Transitions

Diatomic Molecule

■ H₂, CO, CS, SiO
■ Energy Levels

$$r = r_1 + r_2 \quad m_1 r_1 = m_2 r_2$$

$$I = m_1 r_1^2 + m_2 r_2^2 = \frac{m_1 m_2}{m_1 + m_2} (r_1 + r_2)^2 = \mu r^2$$

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

$$H = \frac{I\omega^2}{2}$$

$$H = \frac{L^2}{2I} \quad L = I\omega$$

$$|L^2| = \hbar^2 J(J+1) \quad \textit{quantization}$$

$$E_J = \frac{\hbar^2 J(J+1)}{2I} = hBJ(J+1)$$

$$E_{J+1} - E_J = hB(J+1)(J+2) - hBJ(J+1) = 2hB(J+1)$$

$$\nu = \frac{\Delta E}{h} = 2B(J+1)$$

$$\Delta J = \pm 1$$

Radio Astronomy - Molecular Transitions

Diatomic Molecule

■ Radiative Transfer

$$\frac{dI_\nu}{ds} = -\kappa_\nu I_\nu + \varepsilon_\nu$$

$$I_\nu(s_0) = I_\nu(0)e^{-\tau_\nu(s_0)} + e^{-\tau_\nu(s_0)} \int_0^{s_0} \varepsilon_\nu e^{\tau_\nu(s)} ds$$

$$\tau_\nu(s) = \int_0^s \kappa_\nu ds$$

$$\varepsilon_\nu = \frac{h\nu A\phi(\nu)n_u}{4\pi} \quad \kappa_\nu = \frac{\lambda^2 A\phi(\nu)}{8\pi} \left(\frac{g_u}{g_l} n_l - n_u \right)$$

$$\int \phi(\nu) d\nu \equiv 1$$

$$I_0(s_0) = I_0(0)e^{-\tau_0(s_0)} + \left(\frac{\varepsilon_0}{\kappa_0} \right) (1 - e^{-\tau_0(s_0)})$$

■ Line Intensities

$$\Delta I_0 \equiv I_0(s_0) - I_0(0) = \left[\frac{\varepsilon_0}{\kappa_0} - I_0(0) \right] (1 - e^{-\tau_0(s_0)})$$

Radio Astronomy - Molecular Transitions

Diatomic Molecule

Column Density

$$\tau_0 = \frac{2\sqrt{\ln 2}\lambda^2 A}{8\pi^{(3/2)}\delta\nu} \left(\frac{g_u}{g_l} N_l - N_u \right) \quad N_l = \int_0^{s_0} n_l ds$$

$$\tau_0 = \frac{2\sqrt{\ln 2}\lambda^2 A}{8\pi^{(3/2)}\delta\nu} \frac{g_u}{g_l} N_l (1 - e^{-h\nu/kT_e})$$

$$\frac{n_u}{n_l} = \frac{g_u}{g_l} e^{-h\nu_0/kT_e}$$

Partition Function

$$\frac{N_l}{N} = \frac{g_l e^{-E_l/kT_k}}{Q}$$

$$Q = \sum_{J, \dots} g_{J, \dots} e^{-(E_{J, \dots}/kT_k)}$$

$$Q = \sum_J (2J + 1) e^{-J(J+1) \frac{hB}{kT_k}} \quad \begin{aligned} g_J &= (2J + 1) \\ E_J &= J(J + 1)hB \end{aligned}$$

$$Q = \frac{kT_k}{hB} \quad hB \ll kT_k$$

Radio Astronomy - Molecular Transitions

Symmetric Top Molecule

■ CH₄, CH₃CN, NH₃

■ Energy Level

$$H = \frac{I_x \omega_x^2}{2} + \frac{I_y \omega_y^2}{2} + \frac{I_z \omega_z^2}{2} = \frac{L_x^2}{2I_x} + \frac{L_y^2}{2I_y} + \frac{L_z^2}{2I_z}$$

$$I_x = I_y = I_B \quad I_z = I_C \quad L^2 = L_x^2 + L_y^2 + L_z^2$$

$$H = \frac{L^2}{2I_B} + L_z^2 \left(\frac{1}{2I_C} - \frac{1}{2I_B} \right)$$

$$|L^2| = \hbar^2 J(J+1) \quad |L_z| = \hbar^2 K$$

$$E_{J,K} = J(J+1) \frac{\hbar^2}{2I_B} + K^2 \left(\frac{\hbar^2}{2I_C} - \frac{\hbar^2}{2I_B} \right)$$

$$B = \frac{h}{8\pi^2 I_B} \quad C = \frac{h}{8\pi^2 I_C}$$

$$E_{J,K} = hBJ(J+1) + h(C-B)K^2$$

$$\Delta J = 0, \pm 1, \Delta K = 0$$

Radio Astronomy - Molecular Transitions

Asymmetric Top Molecule

■ HCOOH, C₂H₅CN

■ Energy Levels

$$H = \frac{I_x \omega_x^2}{2} + \frac{I_y \omega_y^2}{2} + \frac{I_z \omega_z^2}{2} = \frac{L_x^2}{2I_x} + \frac{L_y^2}{2I_y} + \frac{L_z^2}{2I_z}$$

$$H = \frac{4\pi^2 A}{h} L_x^2 + \frac{4\pi^2 B}{h} L_y^2 + \frac{4\pi^2 C}{h} L_z^2$$

Prolate : $A > B = C$ *Oblate* : $A = B > C$

$$E = E_{J, K_-, K_+}$$

■ Selection Rules

$$\Delta J = 0, \pm 1$$

Radio Astronomy - Molecular Transitions

Additional Considerations

- Elastic Diatomic Molecules

$$E_J = hBJ(J + 1) - hD[J(J + 1)]^2$$

- NH₃: inversion

- H₂O, H₂CO: ortho v.s. para

- CH₃OH: internal rotor

- Unbalanced shell - Electronic Angular Momentum

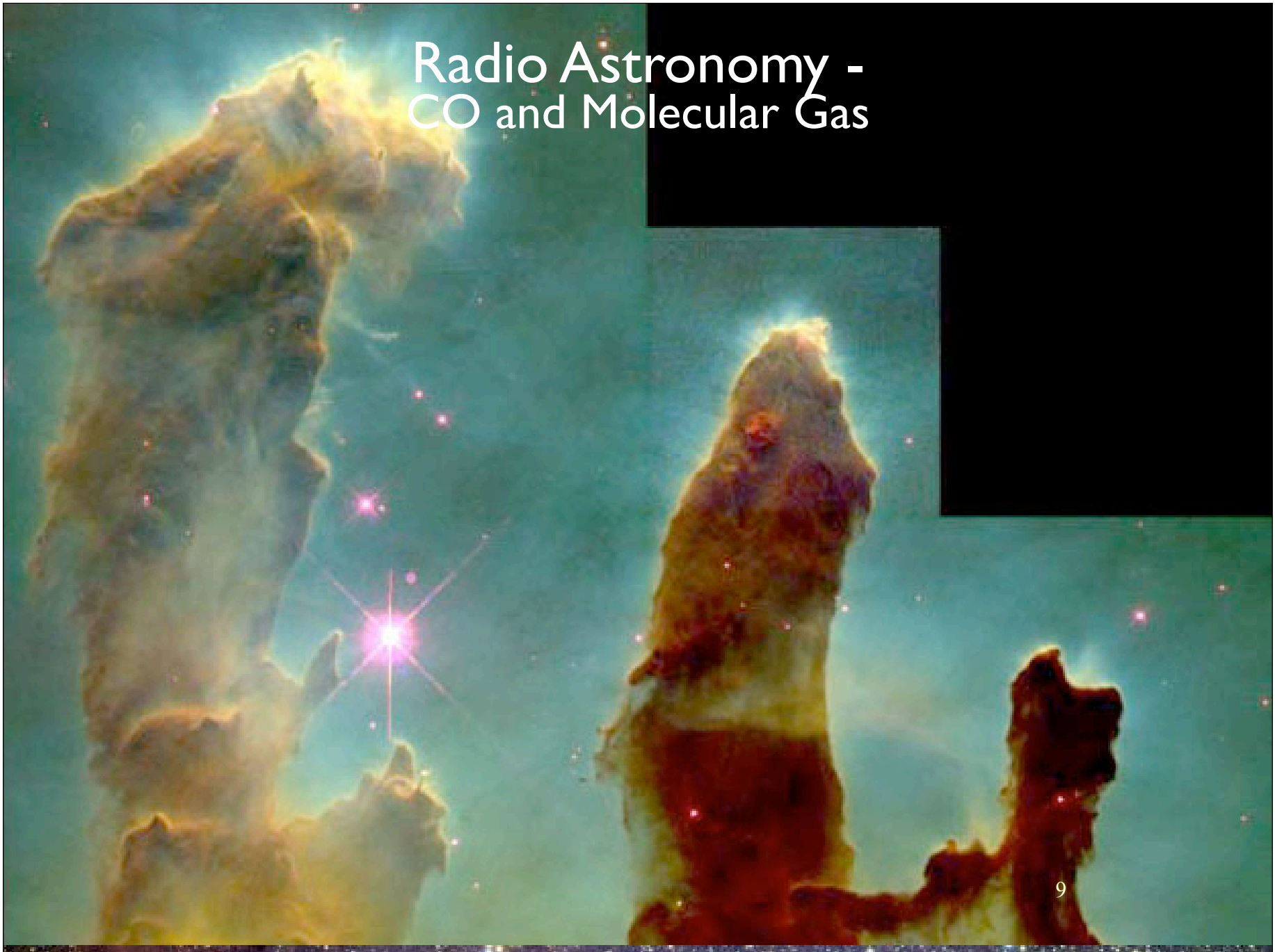
- Quadrupole hyperfine lines, Magnetic Hyperfine lines ...

Radio Astronomy - Molecular Transitions

Applications

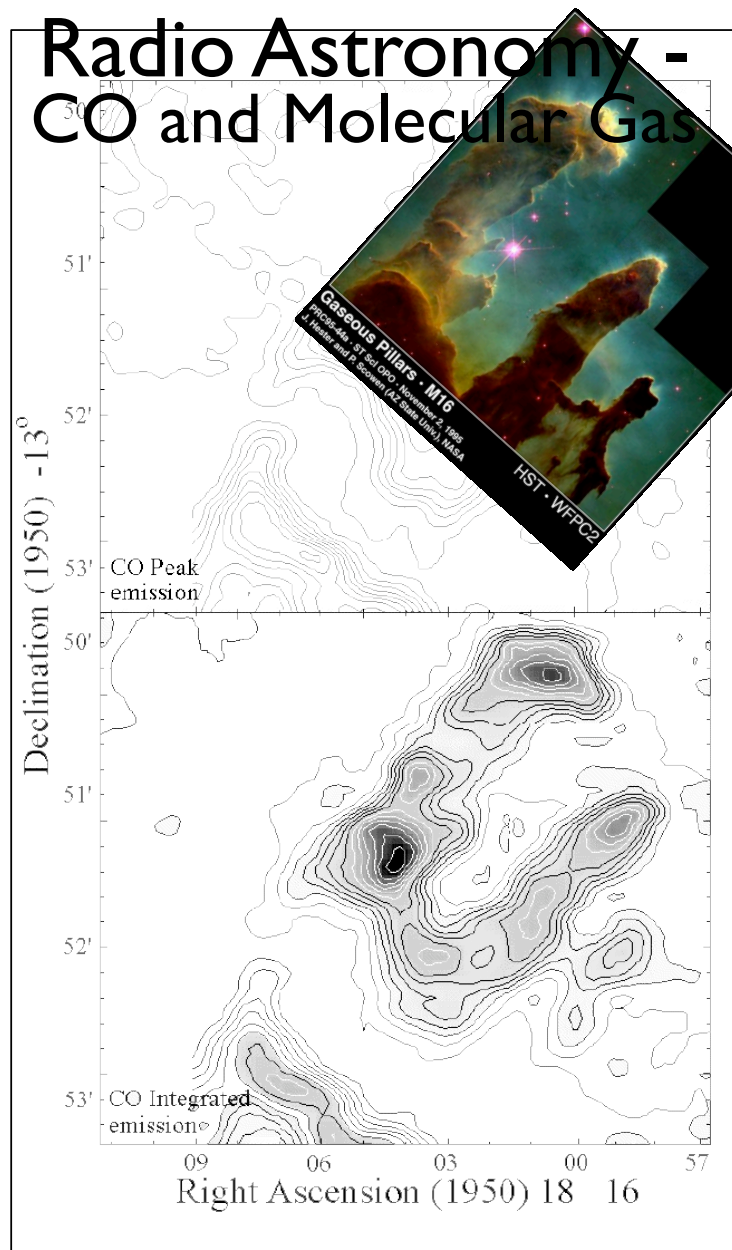
- as Probes of Kinematics & Masses
 - line velocity
 - line width, virial mass
- as Probes of Temperature
 - brightness temperature in optically thin lines
 - line intensity ratios - particularly, K-ladder for symmetric top
- as Probes of Density
 - thermalized emission
- as Probes of Molecular Chemistry itself
 - abundances

Radio Astronomy - CO and Molecular Gas

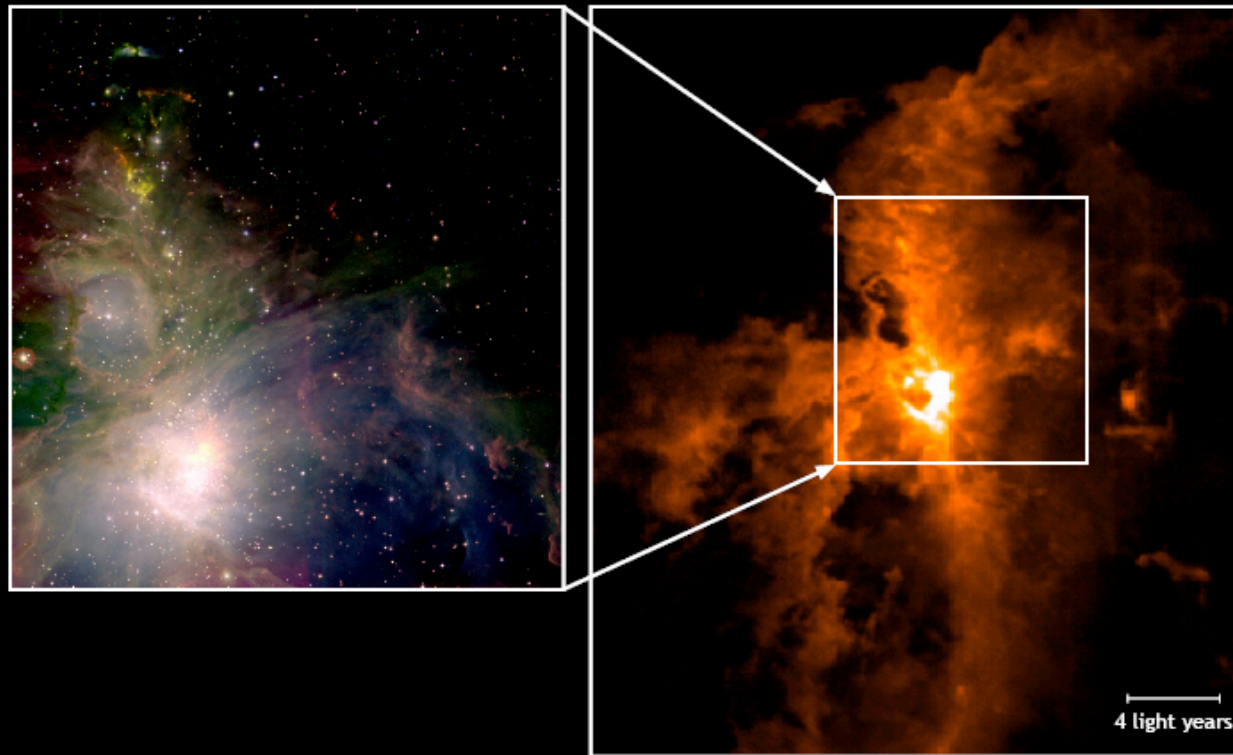


Eagle Nabula (M16)

Radio Astronomy - CO and Molecular Gas



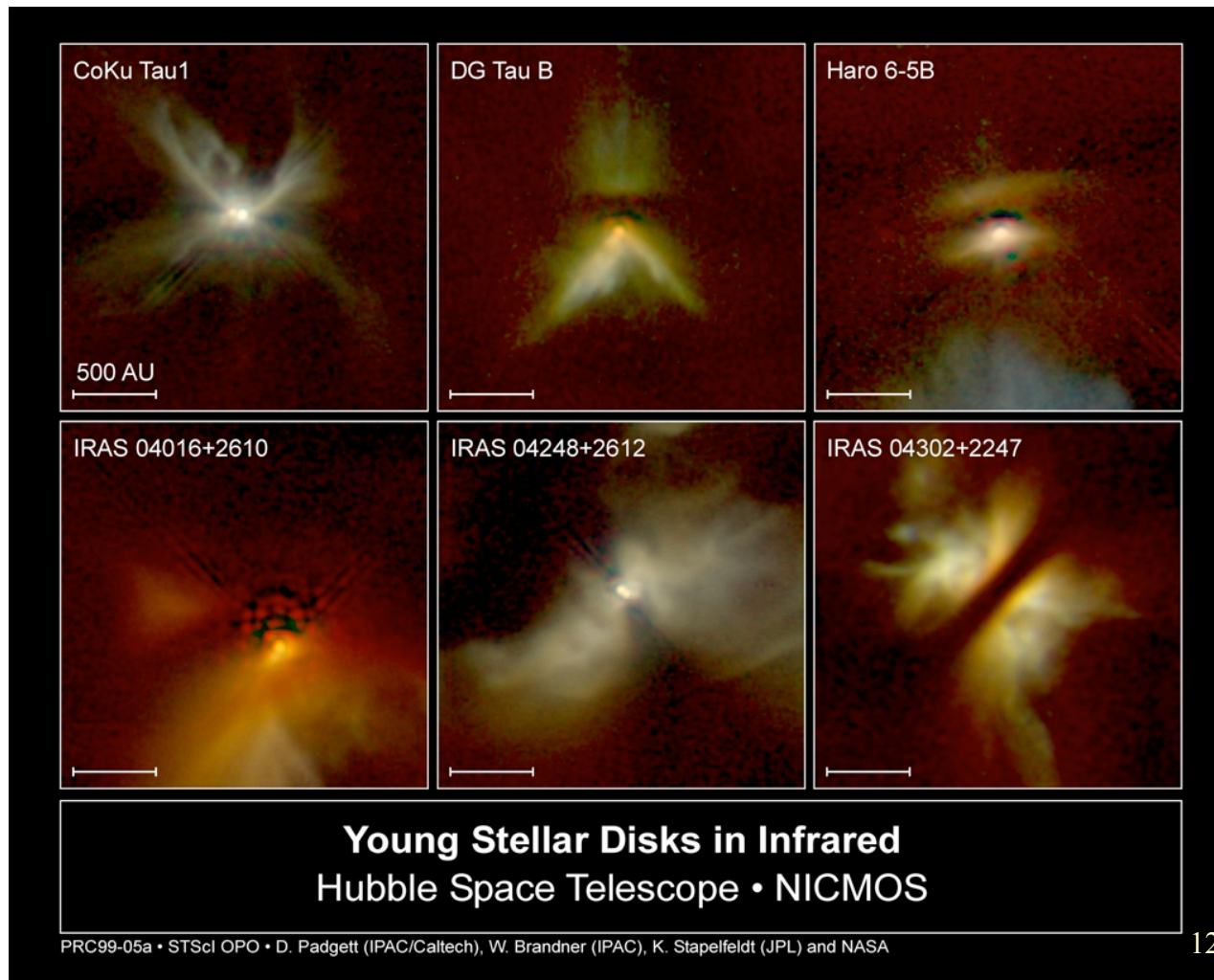
Radio Astronomy - CO and Molecular Gas



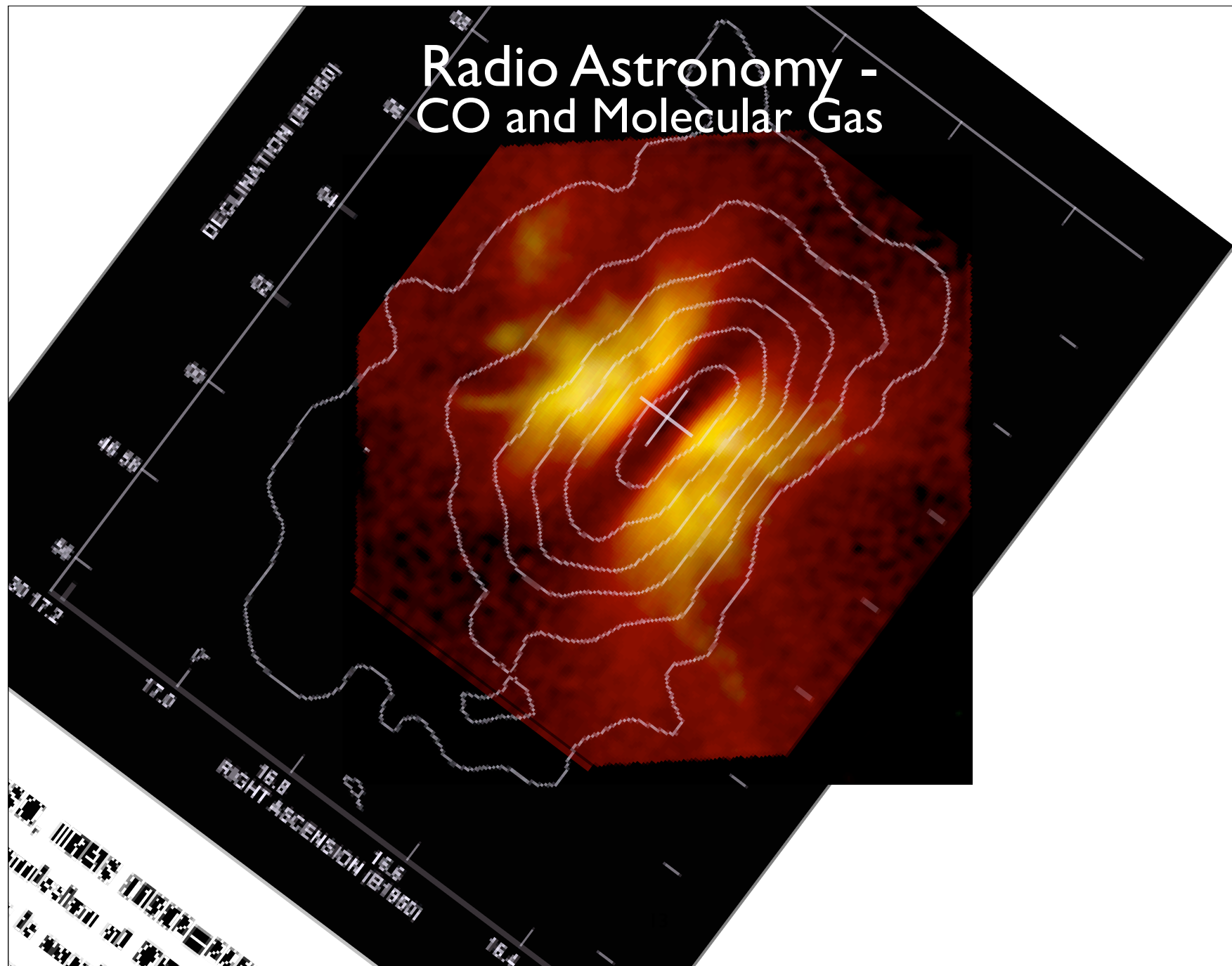
Slicing the Universe with HARP/ACSIS - A New Look at Orion

The image on the right shows Orion obtained with HARP/ACSIS on the James Clerk Maxwell Telescope (JCMT) in the submillimetre wavelength, while the image on the left shows a comparison image obtained with WFCAM on the United Kingdom Infrared Telescope (UKIRT) in the infrared wavelengths. The JCMT submillimetre images shows the dense cloud material from which new stars form. The UKIRT infrared image shows the young stars and the heated gas filaments surrounding them. Both telescopes are operated by the Joint Astronomy Centre in Hilo, Hawai'i. The bar corresponds to four light years, which is equal to the distance from our Sun to the nearest star. (JAC-PR-07-01)

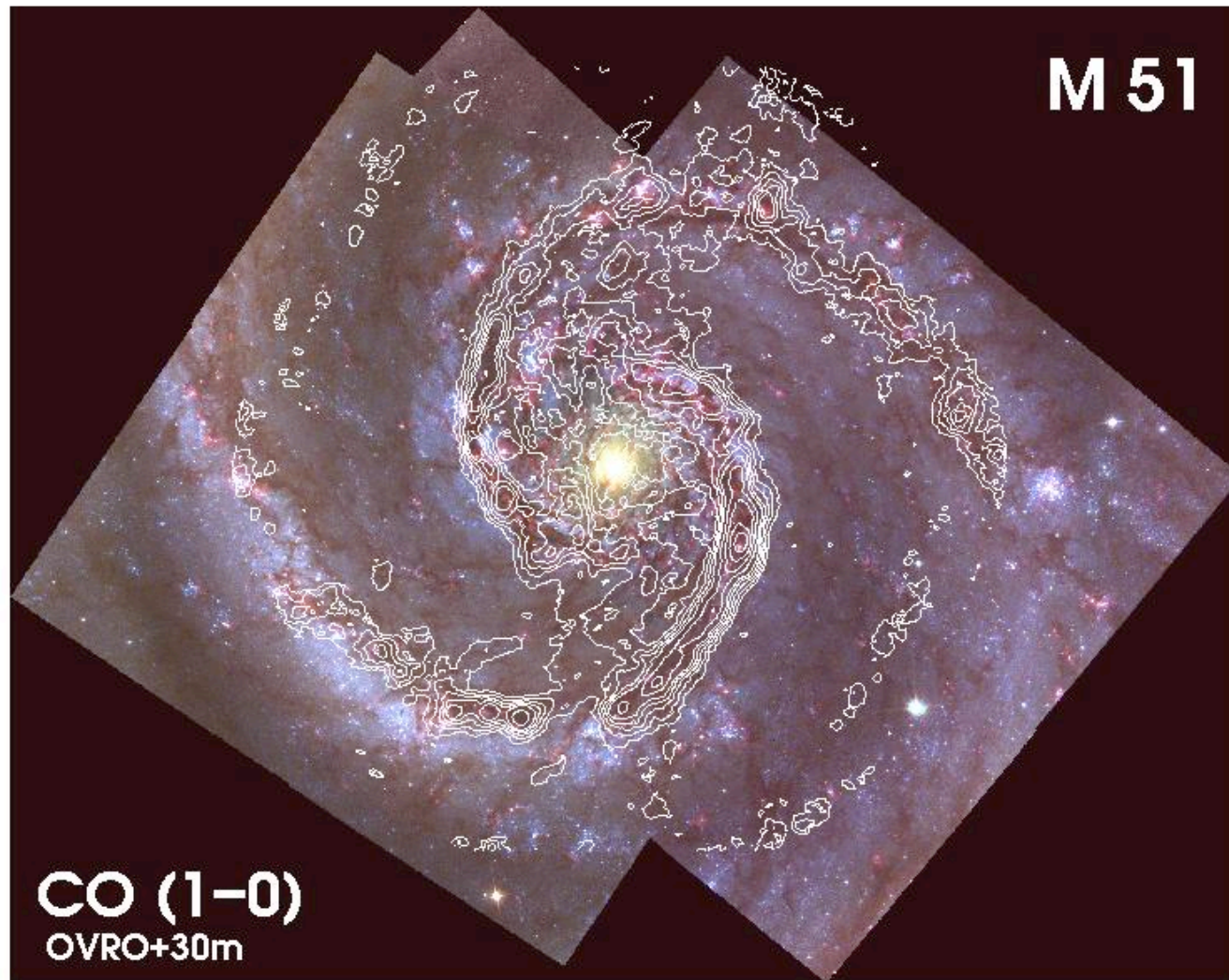
Radio Astronomy - CO and Molecular Gas



Radio Astronomy - CO and Molecular Gas



Radio Astronomy - CO and Molecular Gas



Radio Astronomy - CO and Molecular Gas

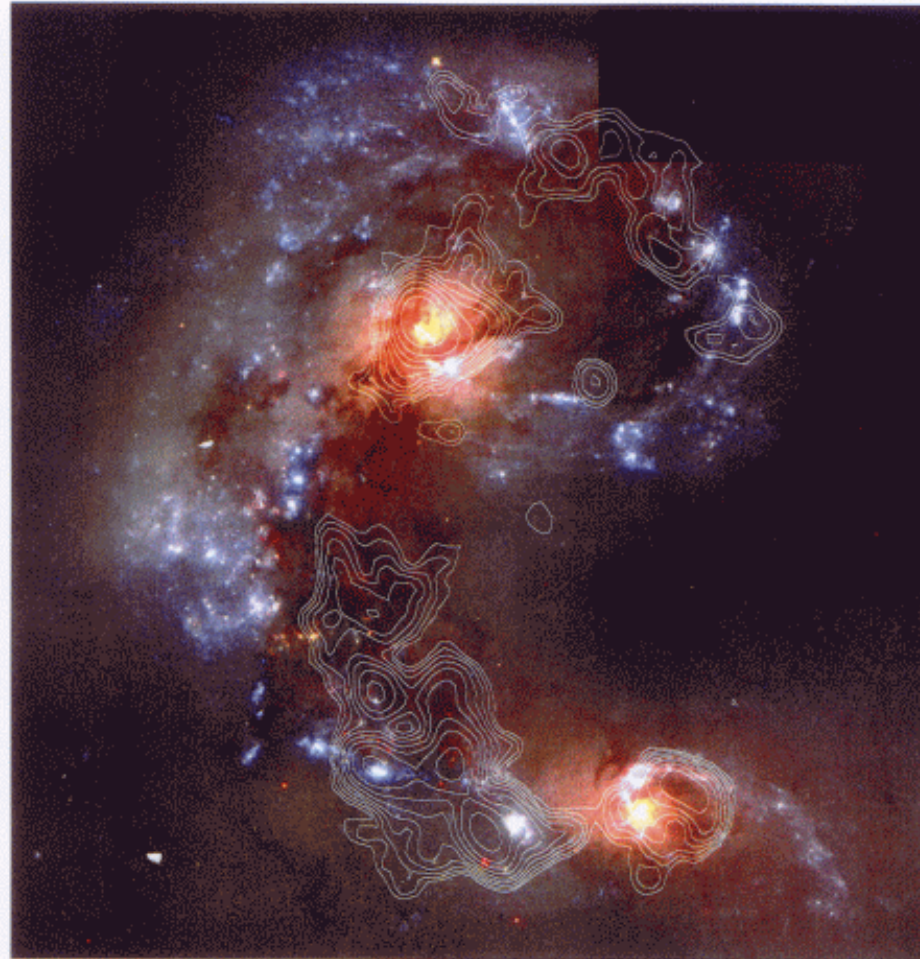
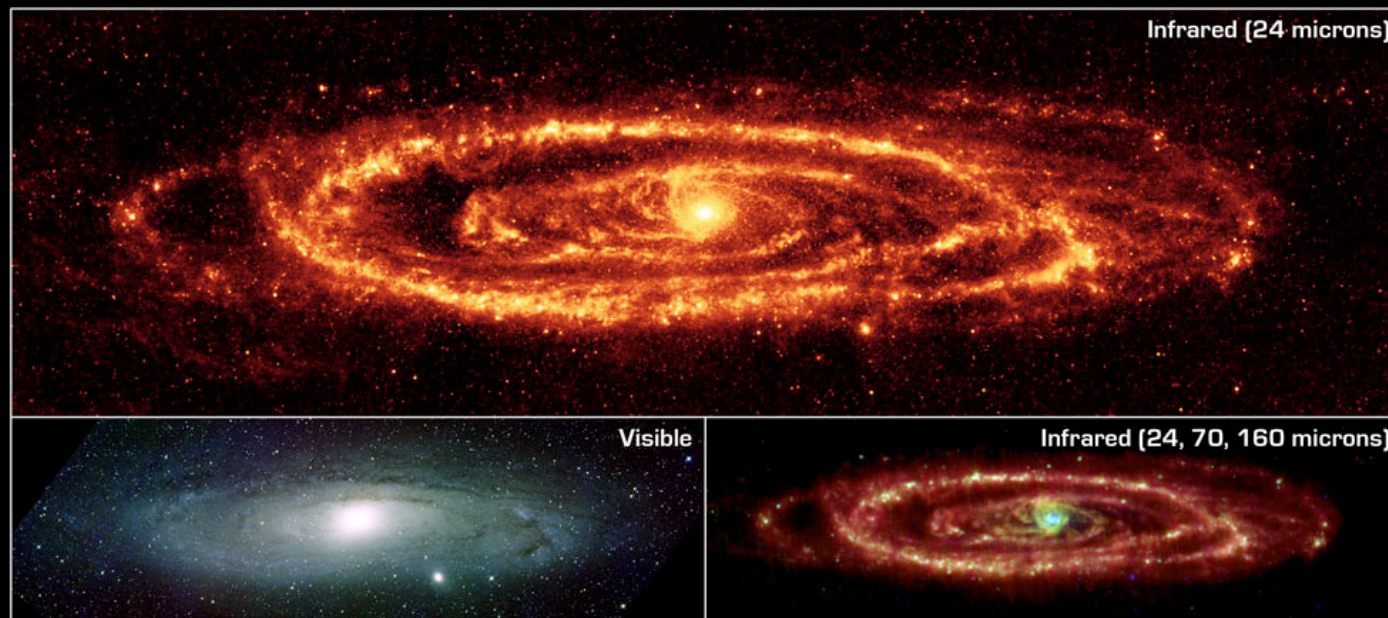


Fig. 4.— Contours of CO(1-0) emission in two colliding galaxies, NGC4038/4039 (the Antennae) overlaid on an HST image. Massive concentrations of molecular gas are located at the nucleus of each galaxy and in the interaction region.

Radio Astronomy - CO and Molecular Gas



Dust in Andromeda Galaxy (M31)

NASA / JPL-Caltech / K. Gordon (University of Arizona)

Spitzer Space Telescope • MIPS

Visible: NOAO

ssc2005-20a

Radio Astronomy - CO and Molecular Gas



Radio Astronomy - Line Emission

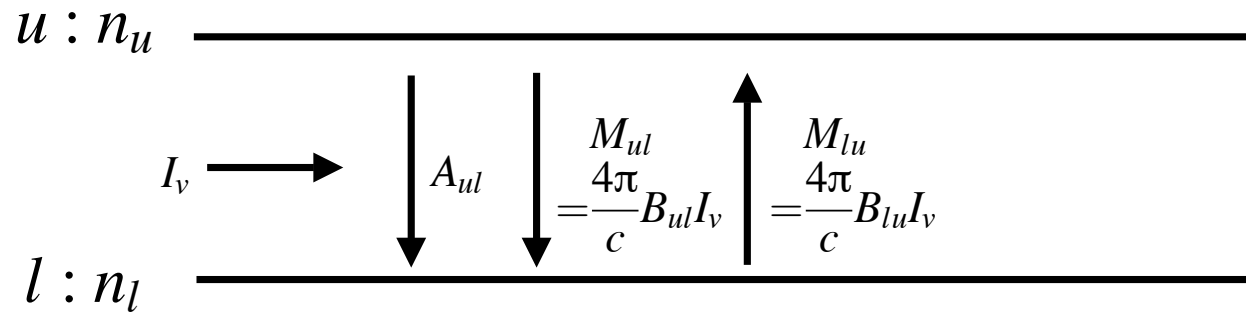
- Masers (Microwave Amplified by Stimulated Emission of Radiation) - non-thermal
 - Theory
 - NH_3 (Gordon et al. 1954)
 - 1964 Nobel Prize Basov, Prokhorov, Townes
 - Cosmic Discovery
 - OH (Weaver et al. 1965)
 - H_2O and many more others
 - Properties
 - compact (e.g. 1×10^4 cm)
 - high brightness temperature (e.g. 1×10^4 - 1×10^{12} K)
 - narrow line width (e.g. 0.2-0.4 km/s)
 - significant polarization
 - time variable

Radio Astronomy - Line Emission

- two-level (radiative)

$$\frac{dI_\nu}{ds} = -\kappa_\nu I_\nu + \varepsilon_\nu$$

$$\frac{dI_\nu}{ds} = \frac{h\nu_0}{4\pi} \left[(n_u B_{ul} - n_l B_{lu}) \frac{4\pi}{c} I_\nu + n_u A_{ul} \right] \phi(\nu)$$



$$\begin{aligned} \kappa_\nu &= \frac{h\nu_0}{c} \left[(n_l B_{lu} - n_u B_{ul}) \phi(\nu) \right] \\ &= \frac{h\nu_0}{c} n_l B_{lu} \left(1 - \frac{g_l n_u}{g_u n_l} \right) \phi(\nu) \\ \varepsilon &= \frac{h\nu_0}{4\pi} n_u A_{ul} \phi(\nu) \end{aligned}$$

Radio Astronomy - Line Emission

- two-level (radiative + collisional)

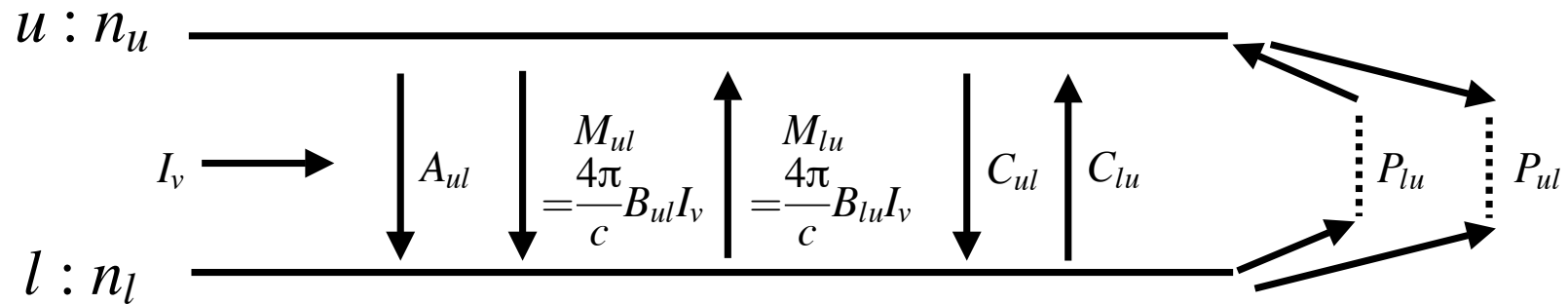
$$\frac{dI_\nu}{ds} = -\kappa_\nu I_\nu + \varepsilon_\nu$$

$$\frac{dI_\nu}{ds} = \frac{h\nu_0}{4\pi} \left[(n_u B_{ul} - n_l B_{lu}) \frac{4\pi}{c} I_\nu + n_u A_{ul} \right] \phi(\nu)$$

$$\kappa_\nu = \frac{h\nu_0}{c} \left[(n_l B_{lu} - n_u B_{ul}) \phi(\nu) \right]$$

$$= \frac{h\nu_0}{c} n_l B_{lu} \left(1 - \frac{g_l n_u}{g_u n_l} \right) \phi(\nu)$$

$$\varepsilon = \frac{h\nu_0}{4\pi} n_u A_{ul} \phi(\nu)$$



$$I = B(T_b) = B(T)$$

LTE

$$\frac{n_u}{n_l} = \frac{g_u}{g_l} e^{-h\nu_0/kT_{ex}} = \frac{g_u}{g_l} e^{-h\nu_0/kT}$$

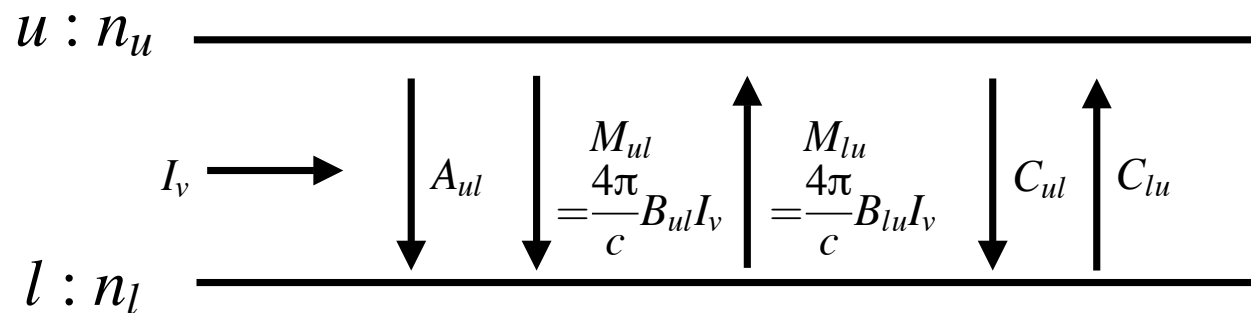
$$T_b = T_{ex} = T_K = T$$

Radio Astronomy - Line Emission

- two-level (radiative + collisional)

$$\frac{dI_\nu}{ds} = -\kappa_\nu I_\nu + \varepsilon_\nu$$

$$\frac{dI_\nu}{ds} = \frac{h\nu_0}{4\pi} \left[(n_u B_{ul} - n_l B_{lu}) \frac{4\pi}{c} I_\nu + n_u A_{ul} \right] \phi(\nu)$$



$$n_u(A_{ul} + C_{ul} + M_{ul}) = n_l(C_{lu} + M_{lu})$$

$$\varepsilon_\nu = \frac{h\nu_0}{4\pi} \frac{n_l(C_{lu} + M_{lu})}{(A_{ul} + C_{ul} + M_{ul})} \phi(\nu)$$

Radio Astronomy - Line Emission

- two-level (radiative + collisional)

$$\epsilon_\nu = \frac{h\nu_0}{4\pi} \frac{n_l(C_{lu} + M_{lu})}{(A_{ul} + C_{ul} + M_{ul})} \phi(\nu)$$

$$\epsilon_\nu = \frac{h\nu_0}{4\pi} \frac{g_u n_l}{g_l} \frac{\left(\frac{I_\nu}{2h\nu_0^3/c^2} + \frac{C_{ul}}{A_{ul}} e^{-h\nu_0/kT_K} \right)}{\left(1 + \frac{I_\nu}{2h\nu_0^3/c^2} + \frac{C_{ul}}{A_{ul}} \right)} \phi(\nu)$$

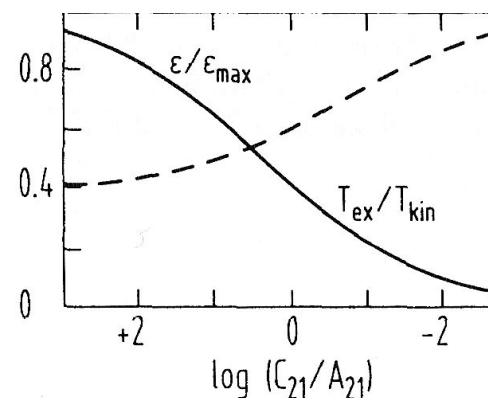
$$\frac{n_u}{n_l} = \frac{g_u}{g_l} e^{-h\nu_0/kT_{ex}}$$

$$\frac{C_{lu}}{C_{ul}} = \frac{g_u}{g_l} e^{-h\nu_0/kT_K}$$

$$\frac{I_\nu}{I_\nu + \frac{2h\nu_0^3}{c^2}} = e^{-h\nu_0/kT_b}$$

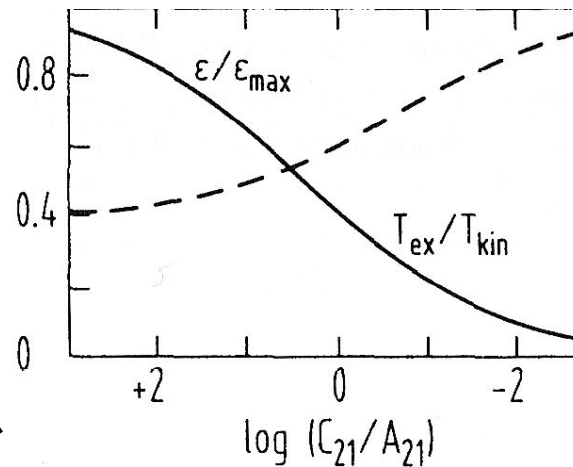
$$T_{ex} = T_K \frac{T_b A_{ul} + T_0 C_{ul}}{T_K A_{ul} + T_0 C_{ul}}$$

$$(T_{ex}, T_K, T_b \gg h\nu_0/k)$$



Radio Astronomy - Line Emission

- two-level (radiative + collisional)



LTE

$$I = B(T_b) = B(T)$$

$$\frac{n_u}{n_l} = \frac{g_u}{g_l} e^{-h\nu_0/kT_{ex}} = \frac{g_u}{g_l} e^{-h\nu_0/kT}$$

$$T_b = T_{ex} = T_K = T$$

**in general
Non-LTE**

$$T_b < T_{ex} < T_K$$

(but,) no masing

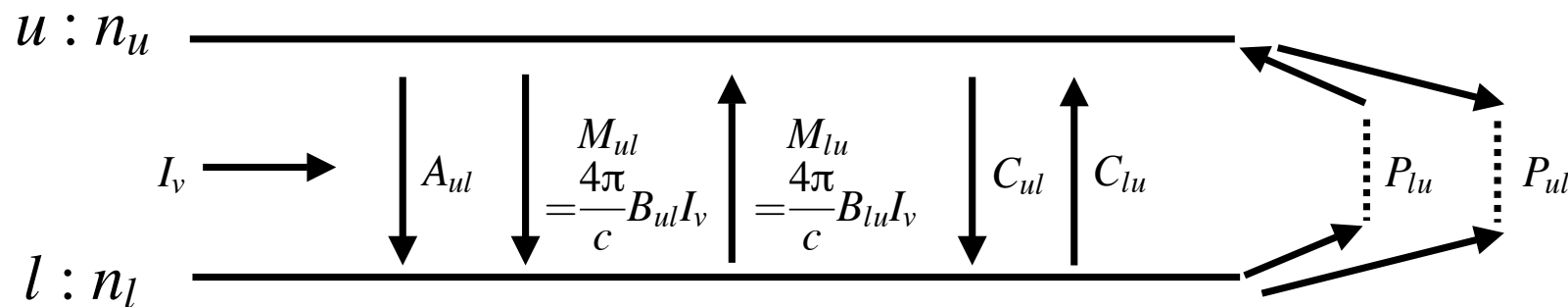
What is critical density?

Radio Astronomy - Line Emission

- three-level (radiative + collisional) or “two-level” + “pumping”

$$\frac{dI_\nu}{ds} = -\kappa_\nu I_\nu + \varepsilon_\nu$$

$$\frac{dI_\nu}{ds} = \frac{h\nu_0}{4\pi} \left[(n_u - n_l) B \frac{4\pi}{c} I_\nu + n_u A \right] \phi(\nu)$$



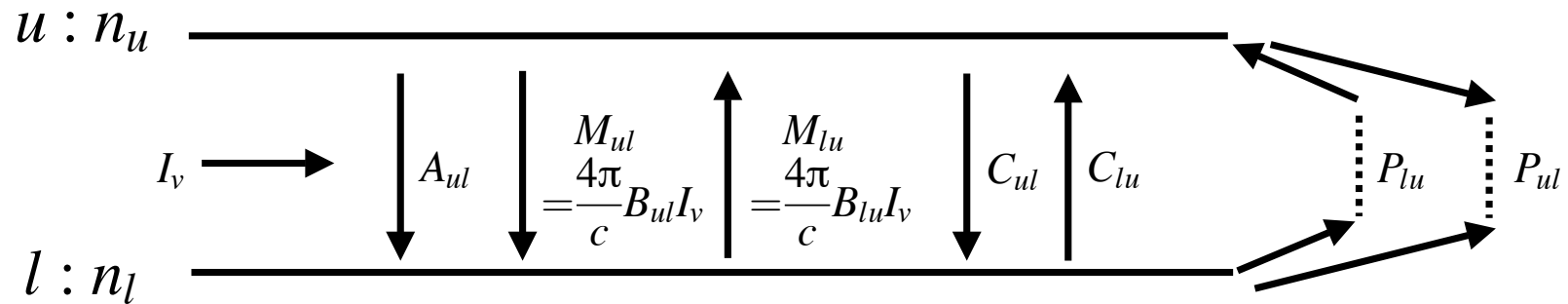
LTE $\frac{n_u}{n_l} = \frac{g_u}{g_l} e^{-h\nu_0/kT_K}$

Radio Astronomy - Line Emission

- three-level (radiative + collisional) or “two-level” + “pumping”

$$\frac{dI_\nu}{ds} = -\kappa_\nu I_\nu + \varepsilon_\nu$$

$$\frac{dI_\nu}{ds} = \frac{h\nu_0}{4\pi} \left[(n_u - n_l) B \frac{4\pi}{c} I_\nu + n_u A \right] \phi(\nu)$$



Non-LTE $n_u(A_{ul} + C_{ul} + M_{ul} + P_{ul}) = n_l(C_{lu} + M_{lu} + P_{lu})$

Radio Astronomy - Line Emission

- Steady State [Statistical Equilibrium]

$$n_u(A_{ul} + C_{ul} + M_{ul} + P_{ul}) = n_l(C_{lu} + M_{lu} + P_{lu})$$

For simplification, assume

$$\begin{aligned} C_{ul} &\approx C_{lu} \approx C \\ M_{ul} &\approx M_{lu} \approx M \\ A &\ll C, A \ll M \end{aligned}$$

$$\frac{n_u}{n_l} = \frac{P_{lu} + M + C}{P_{ul} + M + C}$$

For more simplification, assume $g_u = g_l$

in a special case

$$\begin{aligned} (n_u - n_l)|_{M=C=0} &= (n_u + n_l) \frac{P_{lu} - P_{ul}}{P_{lu} + P_{ul}} \\ &= n \frac{P_{lu} - P_{ul}}{P_{lu} + P_{ul}} \\ &\equiv \Delta n_0 \end{aligned}$$

in a more general case

$$\Delta n = \frac{\Delta n_0}{1 + \frac{2(C+M)}{P}}$$

Radio Astronomy - Line Emission

- Population Inversion

$$\Delta n = \frac{\Delta n_0}{1 + \frac{2(C+M)}{P}}$$

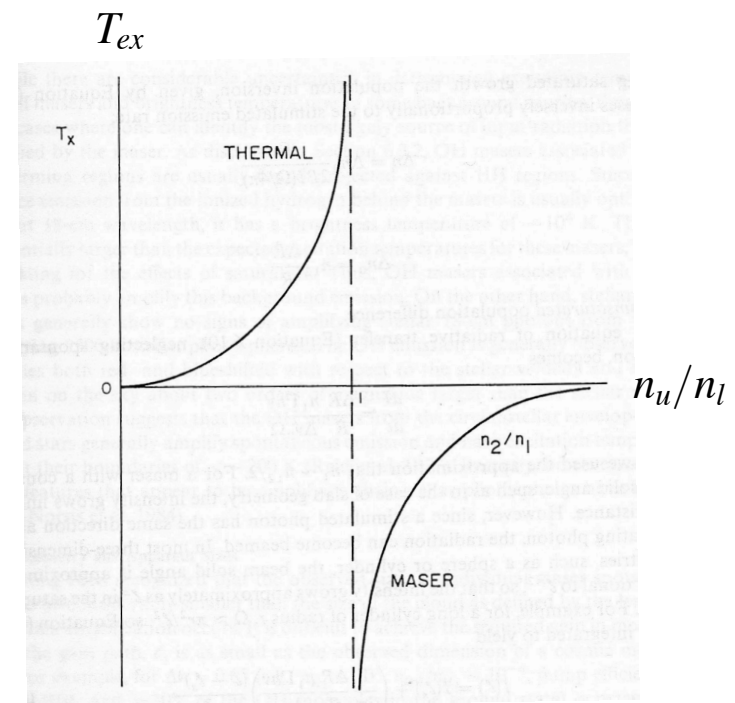
$$\text{If } \Delta n_0 > 0 \Rightarrow \Delta n > 0$$

$$\frac{n_u}{n_l} = e^{-h\nu_0/kT_{ex}} > 1$$

$$\Downarrow$$

$$T_{ex} < 0$$

excitation temperature



Radio Astronomy - Line Emission

- Amplification [in transfer equation]

$$\frac{dI_\nu}{ds} = \frac{\alpha I_\nu}{1 + I_\nu/I_s} + \varepsilon$$

$$\alpha = \frac{h\nu_0}{c} B \frac{\Delta n_0}{1 + \frac{2C}{P}} \phi(\nu)$$

$$I_s = \frac{cP}{2B\Omega_m} \left(1 + \frac{2C}{P}\right)$$

$$\varepsilon = \frac{h\nu_0}{4\pi} n_u A \phi(\nu)$$

Unsaturated
exponential amplification

$$I_{\nu_0} = I_0 e^{\alpha L} + \frac{\varepsilon}{\alpha_0} (e^{\alpha_0 L} - 1)$$

$$T_b = T_c e^{\alpha L} + |T_{ex}| (e^{\alpha_0 L} - 1)$$

Saturated
linear amplification

$$I_{\nu_0} = I_0 + (\alpha_0 I_s + \varepsilon) L$$

Radio Astronomy - Line Emission

- Line width [in transfer equation] : unsaturated case

$$I_{\nu_0} = I_0 e^{\alpha L} + \frac{\epsilon}{\alpha_0} (e^{\alpha_0 L} - 1)$$

$$T_b = T_c e^{\alpha L} + |T_{ex}| (e^{\alpha_0 L} - 1)$$

consider background amplification

$$T_b(\nu) = T_c e^{\alpha(\nu)L}$$

$$\alpha(\nu) = \alpha_0 \exp\left(-\frac{(\nu - \nu_0)^2}{2\sigma_0^2}\right)$$

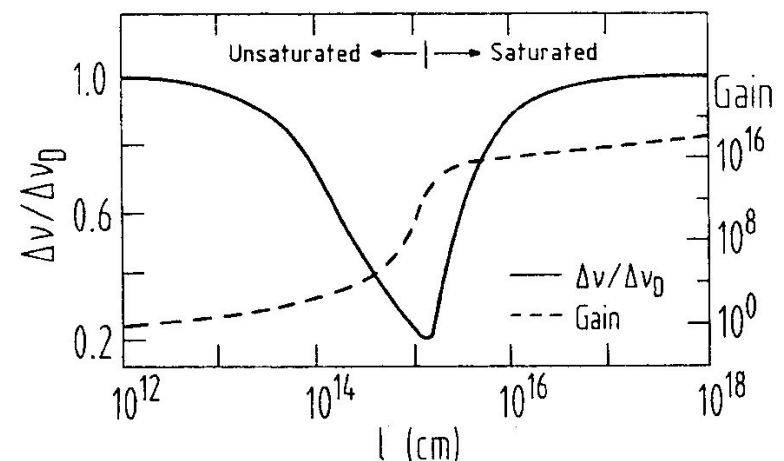
$$T_b(\nu) = T_c e^{\alpha_0 \exp\left(-\frac{(\nu - \nu_0)^2}{2\sigma_0^2}\right)L}$$

$$\alpha(\nu) \approx \alpha_0 \left(1 - \frac{(\nu - \nu_0)^2}{2\sigma_0^2}\right)$$

$$\begin{aligned} T_b(\nu) &= [T_c e^{\alpha_0 L}] e^{-\frac{\alpha_0 L}{2\sigma_0^2} (\nu - \nu_0)^2} \\ &= T_{\nu_0} e^{-\frac{\alpha_0 L}{2\sigma_0^2} (\nu - \nu_0)^2} \end{aligned}$$

line-width and path length

$$\sigma = \frac{\sigma_0}{\alpha_0 L}$$



Radio Astronomy - Masers

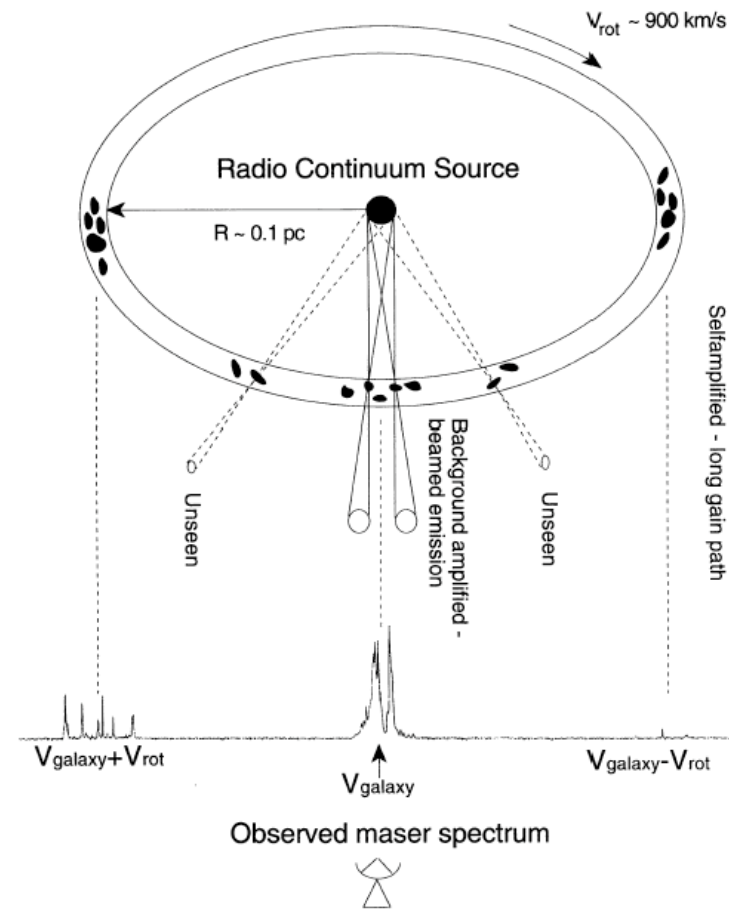
- Masers
 - physical condition required
 - population inversion : still mostly poor understood
 - collisional pumping
 - radiative pumping
 - physical information provided
 - density/temperature (required for pumping)
 - accurate velocity (required for pumping)
 - kinematic information on “small” parcel of gas
 - accurate (relative) positional information
 - proper motion measurements - distance

Radio Astronomy - Masers

- Masers
 - species
 - OH, H₂O, SiO, H₂CO, CH₃OH, HCN, NH₃, H-recomb
 - locations
 - Galactic
 - ISM : star forming regions, HII regions
 - OH, H₂O, SiO, H₂CO, CH₃OH, NH₃, H-recomb
 - stellar : circumstellar envelopes around evolved stars
 - SiO, OH, H₂O, HCN
 - solar : comets
 - Extragalactic
 - “megamaser” ($\sim 10^6$ times brighter than galactic counterparts), or even “gigamaser” for OH
 - H₂O in NGC 4945 (Santos and Lepine 1979)
 - OH in Arp 220 (Baan et al. 1982)
 - circumnuclear disk
 - amplification of radio continuum
 - OH (FIR?), H₂CO (FIR/Radio?), H₂O (collisional?)
 - star formation?

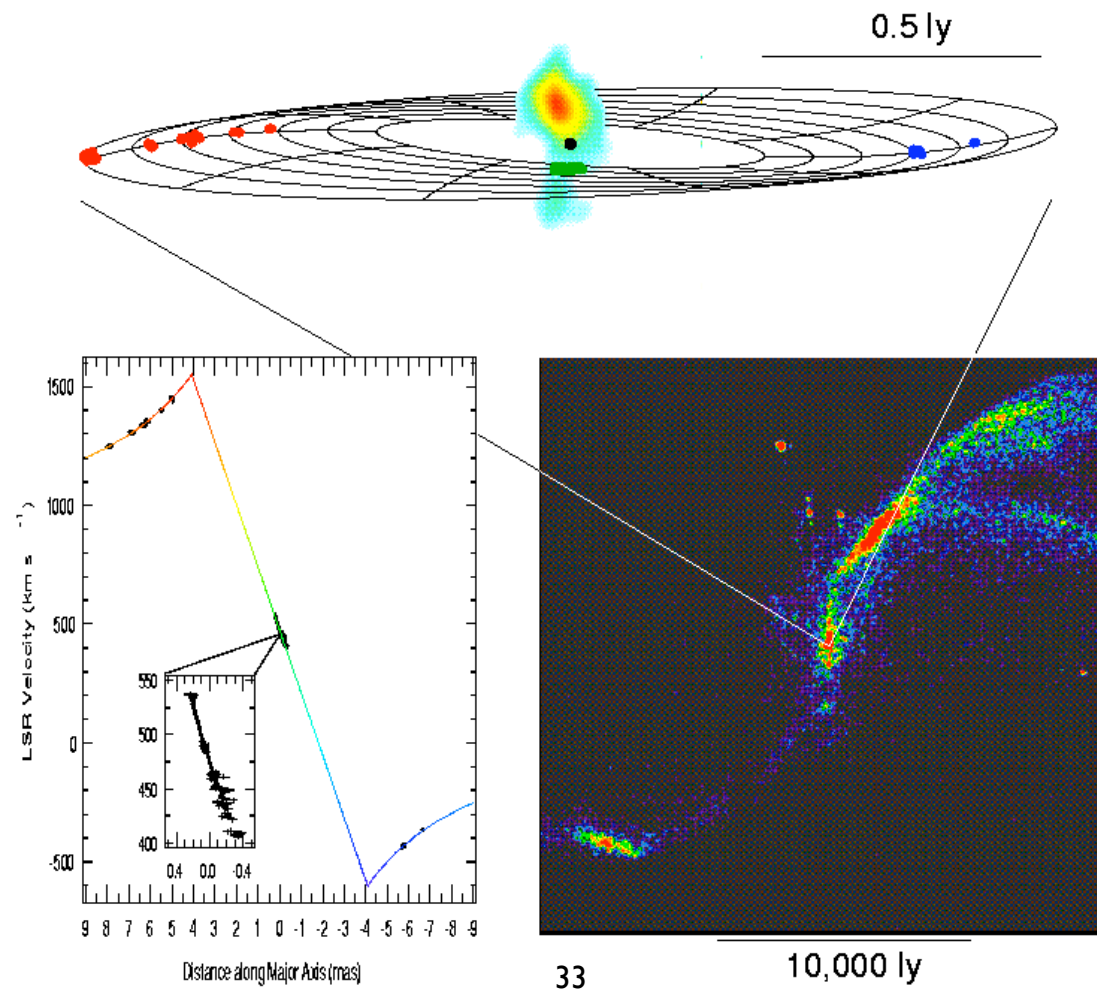
Radio Astronomy - Masers

- Megamasers : SMBH in NGC 4258 (Syfert-2 AGN)



Radio Astronomy - Masers

- Megamasers : SMBH in NGC 4258 (Syfert-2 AGN)



Radio Astronomy - The 21 cm Line

- The 21 cm line
 - first radio wavelengths spectral line detected
 - prediction : van de Hulst and Oort
 - detection : Ewen and Purcell
 - Hyperfine splitting - interaction between the magnetic moment of the proton and the magnetic moment of the orbiting electron at the ground state $1^2S_{1/2}$
- Einstein A ($2.86888 \times 10^{-15}/s$) due to
 - small (magnetic) dipole transition
 - low frequency
- Rest Frequency : 1.420405751786 +/- 0.000000000030 MHz

Radio Astronomy - The 21 cm Line

- The 21 cm line
 - excitation, general called “spin”, temperature

$$\frac{N_u}{N_l} = \frac{g_u}{g_l} \exp\left(\frac{-h\nu_0}{kT_e}\right)$$

$$= 3 \exp\left(\frac{-h\nu_0}{kT_s}\right)$$

- recall

$$T_s = T_K \frac{T_B A_{ul} + T_0 C_{ul}}{T_K A_{ul} + T_0 C_{ul}}$$

$$T_0 = \frac{h\nu_0}{k} = 0.0682K$$

$$= \frac{T_B + y T_K}{1 + y}$$

$$y = \frac{h\nu C_{ul}}{k T_K A_{ul}}$$

Purcell & Field (1956)

$$= \frac{T_K + y' T_B}{1 + y'}$$

$$y' = \frac{k T_K A_{ul}}{h\nu C_{ul}}$$

Kulkarni & Heiles (1988)

Radio Astronomy - The 21 cm Line

- The 21 cm line
 - absorption coefficient, optical depth, and column density

$$\kappa_\nu = \frac{3c^2}{32\pi\nu_{ul}} A_{ul} n_H \frac{h}{kT_s} \phi(\nu)$$

$$d\tau(\nu) = -\kappa_\nu(s, \nu) ds$$

$$d\tau\left(\frac{\nu}{\text{km s}^{-1}}\right) = -5.4873 \cdot 10^{-19} \left(\frac{n_H(s)}{\text{cm}^{-3}}\right) \left(\frac{T_s(s)}{\text{K}}\right)^{-1} \left(\frac{\phi(\nu)}{\text{km}^{-1}\text{s}}\right) d\left(\frac{s}{\text{cm}}\right)$$

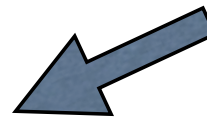
$$\tau(\nu) = \int -\kappa_\nu(s, \nu) ds$$

$$\begin{aligned} \tau\left(\frac{\nu}{\text{km s}^{-1}}\right) &= -5.4873 \cdot 10^{-19} \left(\frac{T_s}{\text{K}}\right)^{-1} \left(\frac{\phi(\nu)}{\text{km}^{-1}\text{s}}\right) \int \left(\frac{n_H(s)}{\text{cm}^{-3}}\right) d\left(\frac{s}{\text{cm}}\right) \\ &= -5.4873 \cdot 10^{-19} \left(\frac{T_s}{\text{K}}\right)^{-1} \left(\frac{\phi(\nu)}{\text{km}^{-1}\text{s}}\right) \frac{N_H}{\text{cm}^{-2}} \end{aligned}$$

$$\begin{aligned} \int \tau(\nu) d\nu &= -5.4873 \cdot 10^{-19} \left(\frac{T_s}{\text{K}}\right)^{-1} \frac{N_H}{\text{cm}^{-2}} \int \left(\frac{\phi(\nu)}{\text{km}^{-1}\text{s}}\right) d\nu \\ &= -5.4873 \cdot 10^{-19} \left(\frac{T_s}{\text{K}}\right)^{-1} \frac{N_H}{\text{cm}^{-2}} \end{aligned}$$

$$\Downarrow$$

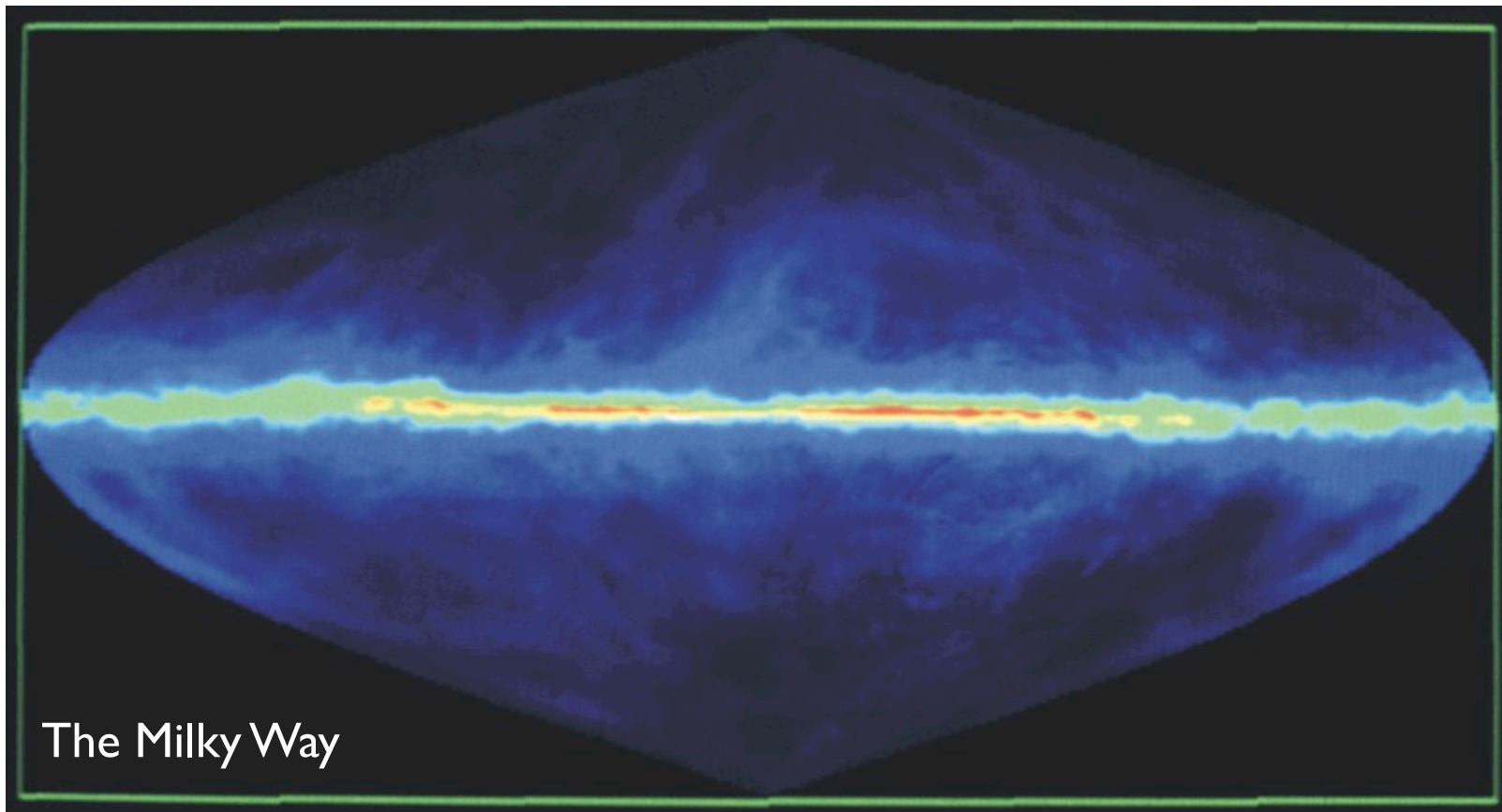
$$\frac{N_H}{\text{cm}^{-2}} = 1.8224 \cdot 10^{18} \left(\frac{T_s}{\text{K}}\right) \int \tau(\nu) d\frac{\nu}{\text{km s}^{-1}}$$



$$\begin{aligned} \tau &= -\ln\left(1 - \frac{T_L}{T_s - T_{BG}}\right) \\ T_s &\gg T_{BG} \\ &\approx -\ln\left(1 - \frac{T_L}{T_s}\right) \\ T_L &\ll T_s \text{ or } \tau \ll 1 \\ &\approx \frac{T_L}{T_s} \end{aligned}$$

Radio Astronomy - The 21 cm Line

- HI Clouds and the Galactic Structure



Radio Astronomy - The 21 cm Line

- HI Clouds and the Galactic Structure

- Interstellar Medium (ISM)

- gas heating and cooling

$$\frac{d}{dt} \left(\frac{3}{2} N k T_k \right) = \Gamma - \Lambda$$

- heating (ionization by radiation, high-energy particles [cosmic rays], collisions of gas clouds or streams)

- cooling (heavy element spectral emission)

- four component model (in pressure balance?) - the swiss cheese picture (with undetermined filling factors)

- CNM (cold neutral medium)

- narrow 21 cm absorption

- $T < 50\text{K}$, $n > 1 - 10 \text{ cm}^{-3}$

- WNM (warm neutral medium)

- ubiquitous wide 21 cm emission

- $T > 200\text{K}$

- WIM (warm ionized medium)

- widespread H α emission

- $T \sim 10^4\text{K}$

- HIM (hot ionized medium)

- diffuse soft X-ray emission and O VI absorption

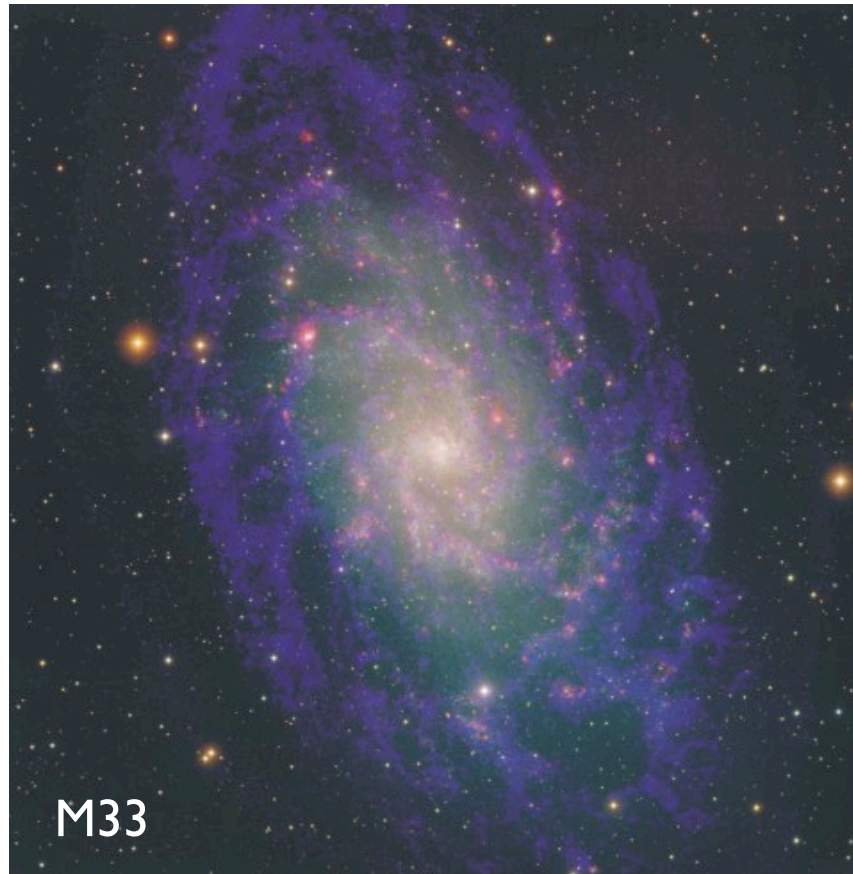
- $T \sim 10^6\text{K}$

Radio Astronomy - The 21 cm Line

- HI in the extragalactic studies
 - extragalaxies
 - major tool for kinematic studies, such as rotation curve since :
 - tracing relative more extended regions since :
 - HI - neutral atomic gas
 - low excitation/density requirement
 - spectral (line) feature - kinematic (velocity) probe
 - findings :
 - HI distribution not centered
 - flat rotation curve
 - tidal interaction features

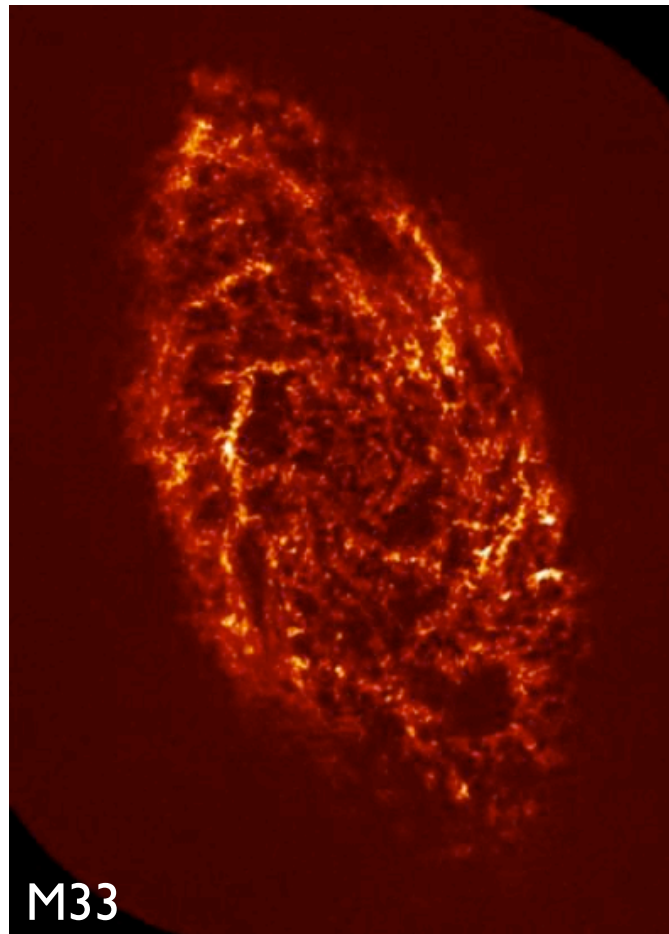
Radio Astronomy - The 21 cm Line

- HI in the extragalactic studies



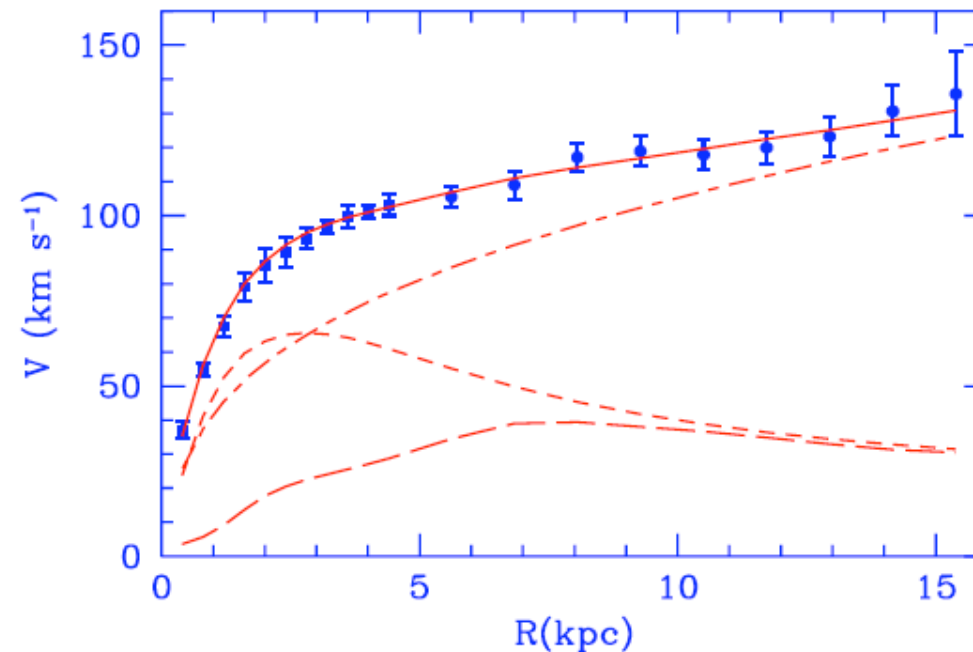
Radio Astronomy - The 21 cm Line

- HI in the extragalactic studies



Radio Astronomy - The 21 cm Line

- HI in the extragalactic studies



M33 rotation curve (points) compared with the best fit model (continuous line). Also shown the halo contribution (dashed-dotted line), the stellar disk (short dashed line) and the gas contribution (long dashed line) [Corbelli and Salucci 2000, MNRAS, 311, 441]

Radio Astronomy - The 21 cm Line

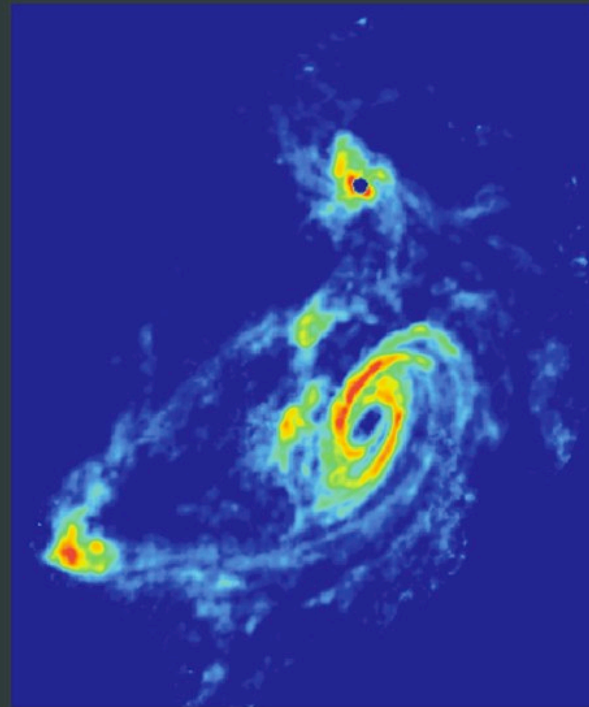
- HI in the extragalactic studies

TIDAL INTERACTIONS IN M81 GROUP

Stellar Light Distribution



21 cm HI Distribution



Radio Astronomy - The 21 cm Line

- HI in the extragalactic studies

